

B.Tech. Degree V Semester Examination, November 2008**ME 505 POWER PLANT ENGINEERING**

(2006 Scheme)

Time: 3 Hours

Maximum Marks: 100

PART A(Answer ALL questions)

(8 x 5)

- I
- Define the following terms:
 - Load factor
 - Diversity factor
 - List the factors which should be considered while designing a power plant.
 - What is a surge tank? Explain with a neat diagram.
 - State the advantages and disadvantages of a gas turbine power plant.
 - Explain the working of an electrostatic precipitator.
 - List down some safety measures for nuclear power plants.
 - Explain the principle of open cycle OTEC system.
 - Write short notes on:
 - MCFC
 - SOFC

PART B

- II A power station has to supply load as follows:

Time (Hours)	0 – 6	6 – 12	12 – 14	14 – 18	18 – 24
Load (MW)	45	135	90	150	75

- Draw the load curve.
- Draw load duration curve.
- Choose suitable generating units to supply the load.
- Calculate the load factor.
- Calculate the plant capacity factor.

(15)

OR

- III A power station has the installed capacity of 180MW. Calculate the cost of generation. Other data pertaining to power station are given below.

Capital cost	= Rs.300 x 10 ⁶
Rate of interest and depreciation	= 18%
Annual cost of fuel, salaries and tax	= Rs..36 x 10 ⁶
Load factor	= 0.4

(15)

- IV List the essential components of a diesel power plant and explain them with neat sketches.

(15)

OR

- V
- Describe with a neat diagram a closed cycle gas turbine power plant.
 - What is a spill way? Briefly explain with a neat sketch.

(10)

(5)

- VI Give the layout of a modern steam power plant and explain it briefly.

(15)

OR

- VII
- Describe with the help of a neat sketch the construction and working of a Pressurized Water Reactor (PWR). What are its advantages and disadvantages?
 - What is boiling water reactor? How it differ from PWR.

(10)

(5)

- VIII Explain briefly with neat sketches.

- Flat plate collector
- Wind turbine
- PEMFC

(15)

OR

- IX
- Describe the working of a geothermal power plant with a neat diagram.
 - Explain the working of thermionic converter with a neat sketch.

(8)

(7)

